
HEPA-Small: Cross-Domain Transfer at 25M Parameters. HEPA Scales.

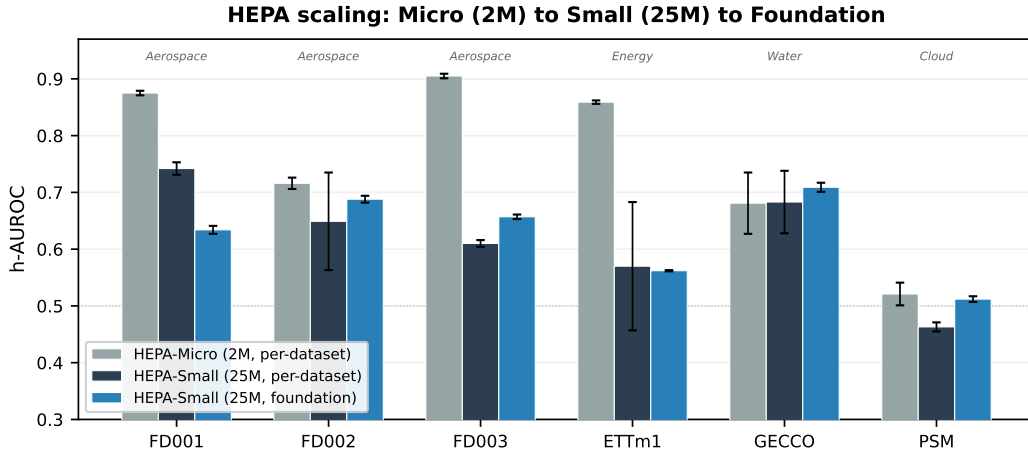


Figure 1: **HEPA scales.** h-AUROC across 6 unseen datasets in 4 domains. A single foundation-pretrained HEPA-Small encoder (25M, light blue) generalizes cross-domain without any encoder finetuning, improving over per-dataset pretraining (dark) on 4 of 6 benchmarks with substantially reduced variance. HEPA-Micro (2M, grey) is the original submission baseline.

Abstract

We report scaling results for HEPA (Horizon-conditioned Event Predictive Architecture) [1], a self-supervised JEPA for event prediction in multivariate time series that received Spotlight at FMSD@ICML 2026. Since the original submission at the 2M-parameter micro scale across 14 datasets, we have scaled HEPA to 25M parameters (HEPA-Small) with a channel-independent (CI) tokenization [3] that enables a single encoder to be pretrained across datasets with varying numbers of input channels. A single HEPA-Small encoder, pretrained on a subset of the open-source Chronos-2 corpus [2], transfers to 6 unseen downstream datasets spanning 4 domains, improving over per-dataset pretraining on 4 of 6 benchmarks (mean Δ h-AUROC = +0.008, median +0.033). These results provide early evidence that HEPA’s encoder scales cross-domain and motivate the HEPA-Base scale-up to 200M parameters. This technical report builds directly on the HEPA paper and presents the first foundation-model results.

1 Introduction

HEPA [1] demonstrated that a 2M-parameter causal JEPA, trained per-dataset, matches or exceeds supervised baselines on event prediction across 14 datasets spanning 11 domains. The architecture separates a *context encoder* (which learns general temporal representations) from a *lightweight predictor* (which is finetuned for the downstream task), with the hypothesis that the encoder can be pretrained once and reused.

Table 1: HEPA model conditions evaluated in this report, plus the prospective HEPA-Base target for Stage 1.

Condition	Tokenization	Encoder	Pretraining	Status
HEPA-Micro	Fused	2.0M	Per-dataset	Original submission
HEPA-Small (per-dataset)	Channel-indep.	25.2M	Per-dataset	This report
HEPA-Small (foundation)	Channel-indep.	25.2M	Chronos-2 corpus	This report
HEPA-Base (foundation)	Channel-indep.	200M	Large mixed corpus	Stage 1 target

Two questions remained open at submission:

1. Does the architecture scale beyond 2M parameters?
2. Can a single pretrained encoder generalize across domains?

We provide early affirmative evidence for both. HEPA-Small (25.2M encoder parameters) is pretrained on the Chronos-2 corpus [2] and evaluated on 6 datasets spanning 4 domains, none seen during pretraining. Foundation pretraining improves over per-dataset pretraining on 4 of 6 benchmarks, with a mean h-AUROC gain of +0.008 and a median gain of +0.033.

2 Method

We refer the reader to Petersen [1] for the full HEPA architecture and training procedure. The key methodological change in this work is enabling pretraining over multiple datasets with **varying numbers of input channels**.

Channel-independent tokenization. The original HEPA uses fused tokenization ($\text{Linear}(C \cdot P, d_{\text{model}})$), which binds the encoder to a fixed channel count C at initialization. This prevents cross-dataset pretraining: when moving to a dataset with a different C , the input projection must be reinitialized, discarding pretrained representations. Following PatchTST [3], we replace this with **channel-independent (CI) tokenization**: each input channel is independently patched ($\text{Linear}(P, d_{\text{model}})$ with patch size $P=16$) and assigned both temporal and channel positional encodings, producing $C \times N$ tokens per sample (where $N = T/P$). A **block-causal attention mask** allows cross-channel attention within each timestep while enforcing causal ordering across time. This makes the encoder agnostic to C : the same weights process any number of input channels without reinitializing any parameters, enabling foundation pretraining across heterogeneous datasets.

Foundation pretraining. We pretrain on a subset of the Chronos-2 corpus [2], which is publicly available as open-source data. We construct pseudo-multivariate samples by grouping $c_{\text{group}} = 8$ univariate series from the same corpus configuration into $(T, C=8)$ samples. The HEPA-Small encoder ($d_{\text{model}} = 256$, $L = 6, 8$ heads, 25.2M parameters) is trained for 5 epochs on 128-token samples (approximately 1.5 hours on a single A10G). We verified that none of the 6 downstream evaluation datasets appear in the pretraining corpus, ensuring no data leakage between foundation pretraining and evaluation. For each downstream dataset, we **freeze the encoder** and train only a fresh predictor and per-horizon sigmoid event head (0.8M parameters, 35–50 epochs).

3 Results

Table 1 summarizes the three evaluated conditions and the prospective HEPA-Base target. Full per-dataset results are in Table 2 (Appendix).

The key observations:

- **Foundation transfer works.** The foundation encoder, pretrained on a general time-series corpus with no domain-specific data, transfers to unseen industrial domains. On PSM (cloud, +0.049), FD003 (aerospace, +0.047), and FD002 (aerospace, +0.039), the gains are substantial.

- **Reduced variance.** Foundation-pretrained models show consistently lower standard deviations across seeds (e.g., ETTm1: ± 0.001 vs. ± 0.113 ; FD002: ± 0.006 vs. ± 0.086), indicating more stable representations. For industrial deployment, this stability may matter as much as the mean improvement.
- **CI tokenization is essential:** with fused tokenization, foundation pretraining *hurt* performance because the input projection must be reinitialized per channel count (see Section 2).
- **FD001 is the sole large regression** (-0.108). Its single operating condition and relatively simple degradation dynamics make it the dataset least likely to benefit from diverse foundation pretraining.

4 Limitations

CI tokenization trades per-dataset accuracy for transferability: at small scale with per-dataset pretraining, CI-HEPA slightly underperforms the original fused tokenization (mean $\Delta = -0.020$), as the $C \times N$ token count creates a harder optimization problem. The pseudo-multivariate pretraining corpus (grouping univariate series) may underrepresent true cross-channel dependencies found in real industrial systems. Foundation pretraining used only 5 epochs due to compute constraints; the pretext loss plateaus rapidly, but longer training with a larger corpus may yield further gains. These limitations are directly addressed by the HEPA-Base data and compute strategy in Stage 1.

5 Reproducibility

Code and data. All code is available in the project repository. Datasets are publicly available: C-MAPSS FD001–FD003 [4], ETTm1 [5], GECCO [6], and PSM [7]. Pretraining uses the Chronos-2 corpus [2].

Compute. Foundation pretraining: approximately 1.5 hours on a single NVIDIA A10G (24GB). Per-dataset finetuning: 5–20 minutes per dataset per seed on the same GPU. Total compute for all results: <10 GPU-hours.

Hyperparameters. Encoder: $d=256$, $L=6$, 8 heads, patch size $P=16$, context length $T=2048$. Pretraining: spectral regularization $\lambda=0.05$, learning rate 3×10^{-4} , AdamW, batch size 64. Finetuning: learning rate 1×10^{-4} , 35–50 epochs, positive-weighted BCE loss. All results averaged over 3 random seeds.

A Detailed Results

Table 2: h-AUROC (mean $\pm$ std, 3 seeds). Bold: better HEPA-Small condition; neither bolded when the difference is within noise.

	FD001	FD002	FD003	ETTm1	GECCO	PSM
HEPA-Micro	.875 $\pm$.00	.716 $\pm$.01	.905 $\pm$.00	.859 $\pm$.00	.681 $\pm$.05	.521 $\pm$.02
Small (per-ds)	.742 $\pm$.01	.649 $\pm$.09	.610 $\pm$.01	.570 $\pm$.11 [†]	.683 $\pm$.06	.463 $\pm$.01
Small (fnd)	.634 $\pm$.01	.688 $\pm$.01	.657 $\pm$.00	.562 $\pm$.00 [†]	.709 $\pm$.01	.512 $\pm$.01
Δ	-.108	+.039	+.047	-.008	+.026	+.049

[†]ETTm1 difference ($\Delta = -0.008$) is within the noise of per-dataset variance (± 0.113).

B Pretraining Corpus

Foundation pretraining uses the real-world subset of the open-source Chronos-2 corpus (autogluon/chronos_datasets on HuggingFace [2]), excluding synthetic corpora (training_corpus_*) and the 14 weatherbench_hourly_* configurations. This yields ~ 50 configurations spanning energy (ERCOT, electricity, solar), traffic (Uber TLC, rideshare), climate (WeatherBench daily/weekly, USHCN), forecasting competitions (M4, M5), web (Wikipedia), and other public collections (Monash repository, FRED).

The vast majority of configurations are univariate (column target); only 3 are natively multivariate: monash_rideshare, monash_temperature_rain, and ushcn_daily. To construct pseudo-multivariate samples, we group $c_{\text{group}}=8$ univariate series from the same configuration into $(T, C=8)$ windows (sampling with replacement for the ~ 4 configurations with fewer than 8 usable series). Each configuration contributes equally per epoch (balanced sampling), preventing large configurations from dominating.

References

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